

Mutual Fund Analytics Capstone Project



Final Project Report

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Internship

Bluestock Fintech – Data Analyst Internship

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Executive Summary

This project presents a comprehensive analysis of mutual fund data using data analytics and visualization techniques. The objective was to evaluate mutual fund performance, identify risk-return patterns, and provide actionable insights for investors.

The project involved an end-to-end data analytics workflow, including data extraction, transformation, and loading (ETL), followed by Exploratory Data Analysis (EDA), performance evaluation, and risk assessment. Key financial metrics such as Sharpe Ratio, Value at Risk (VaR), and Conditional Value at Risk (CVaR) were used to measure fund performance and risk exposure.

An interactive Power BI dashboard was developed to visualize fund performance, category-wise comparisons, and risk metrics. The analysis revealed differences in performance across fund categories and highlighted the importance of balancing returns with risk considerations.

The findings of this project demonstrate how data-driven analysis can support informed investment decisions. The dashboard and analytical insights provide a practical tool for evaluating mutual funds and understanding their risk-return characteristics.

1. Project Objective

The main objectives of this project are:

- Analyze mutual fund performance.
- Evaluate risk and return characteristics.
- Perform Exploratory Data Analysis (EDA).
- Calculate financial metrics such as Sharpe Ratio and Value at Risk (VaR).
- Build an interactive Power BI dashboard.
- Generate recommendations for investors.

2. Dataset Overview

The dataset contains information related to various mutual funds including:

- Fund Name
- Fund Category
- Net Asset Value (NAV)
- Returns
- Risk Measures
- Fund Size
- Expense Ratio
- The dataset was cleaned and transformed before analysis.

3. ETL Process

Extract

The mutual fund dataset was collected and imported into Python.

Transform

Removed missing values.

Handled duplicate records.

Standardized column names.

Converted data types where required.

Load

The cleaned dataset was loaded into analytical notebooks and Power BI for visualization.

4. Exploratory Data Analysis (EDA)

Several analyses were performed to understand the data.

Key Findings

- Equity funds generally provided higher returns.
- Hybrid funds showed balanced risk and return.
- Debt funds had lower volatility.
- Some fund categories significantly outperformed others.

Visualizations used:

- Distribution plots
- Category-wise comparisons
- Return analysis charts
- Risk analysis charts

5. Performance Analysis

The performance of mutual funds was evaluated using various financial metrics.

Sharpe Ratio

The Sharpe Ratio was calculated to measure risk-adjusted returns.

Rolling Sharpe Analysis

Rolling Sharpe Ratio analysis helped identify performance consistency over time.

CAGR Analysis

Compound Annual Growth Rate was used to evaluate long-term growth.

6. Risk Analysis

Risk metrics were calculated to assess potential losses.

Value at Risk (VaR)

VaR estimates the maximum expected loss at a given confidence level.

Conditional Value at Risk (CvaR)

CVaR measures the average loss beyond the VaR threshold.

Findings

- High-return funds often carried higher risk.
- Diversified funds showed lower downside risk.
- Certain categories maintained better risk-adjusted performance.

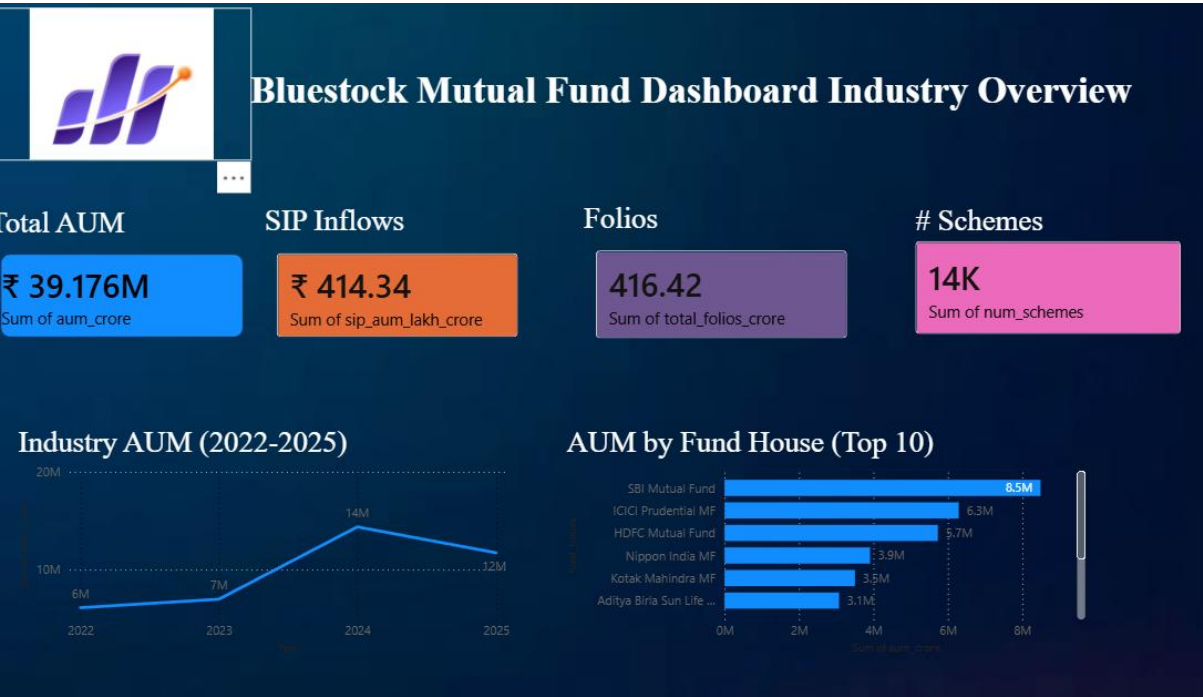
7. Dashboard Development

A Power BI dashboard was developed to visualize key metrics.

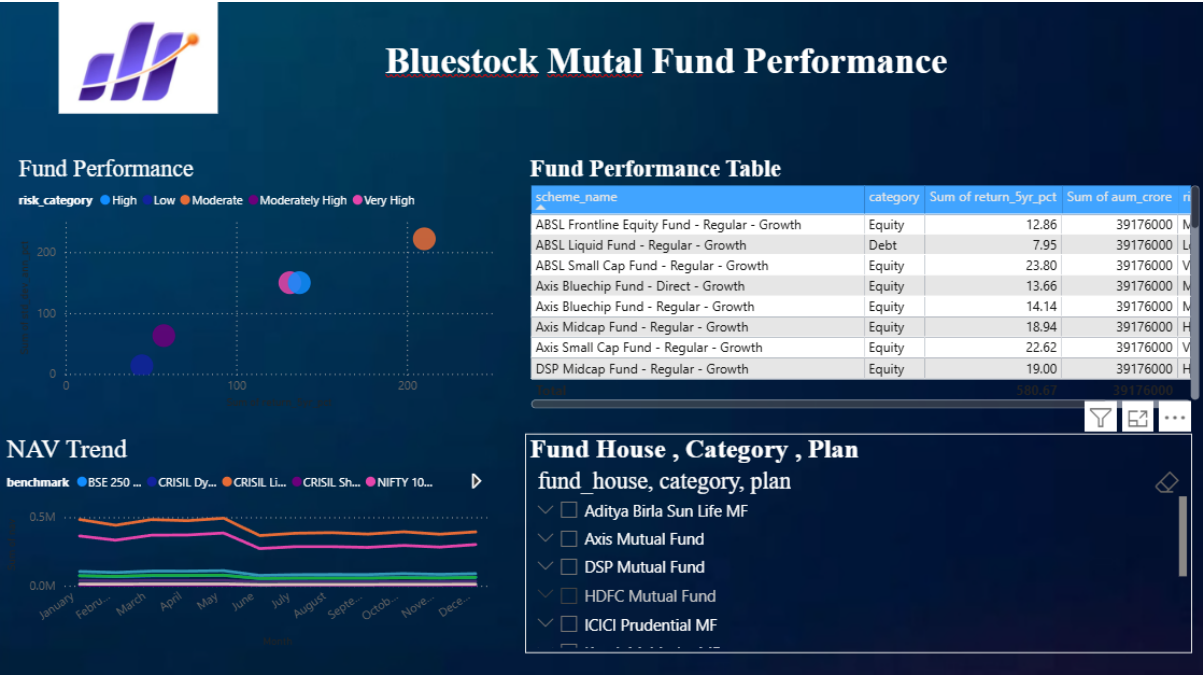
Dashboard features include:

- Fund Performance Overview
- Category Analysis
- Risk Metrics
- Return Comparisons
- Interactive Filters
- Dashboard screenshots are included in the appendix.

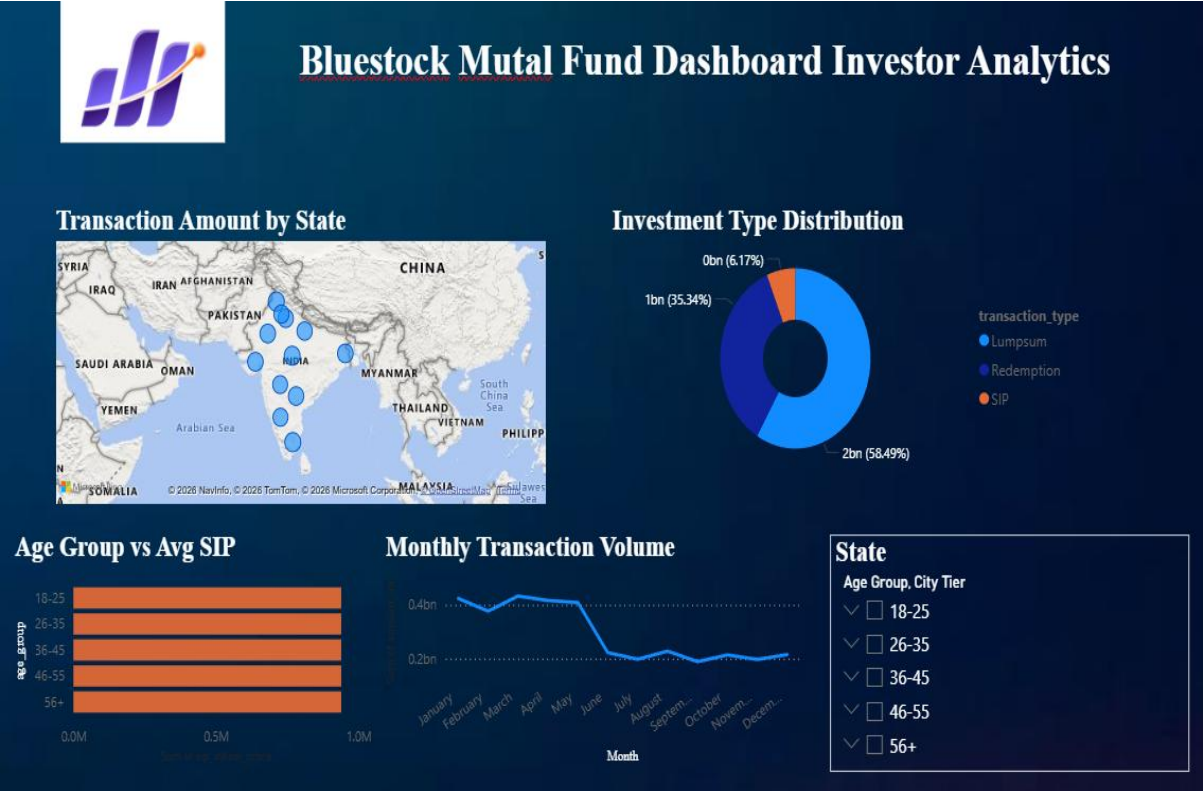
Fund Performance Overview



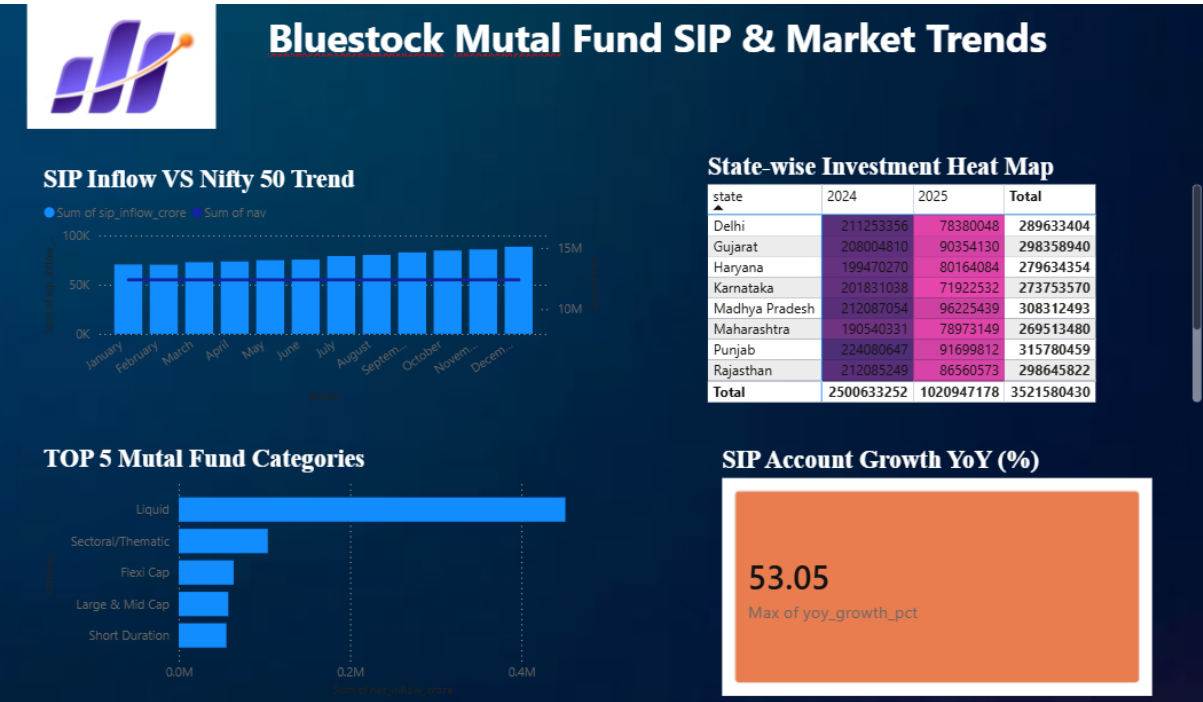
Mutal Fund Performance



Mutal Fund Dashboard Investor Analytics



Fund SIP & Market Trends



Recommendations

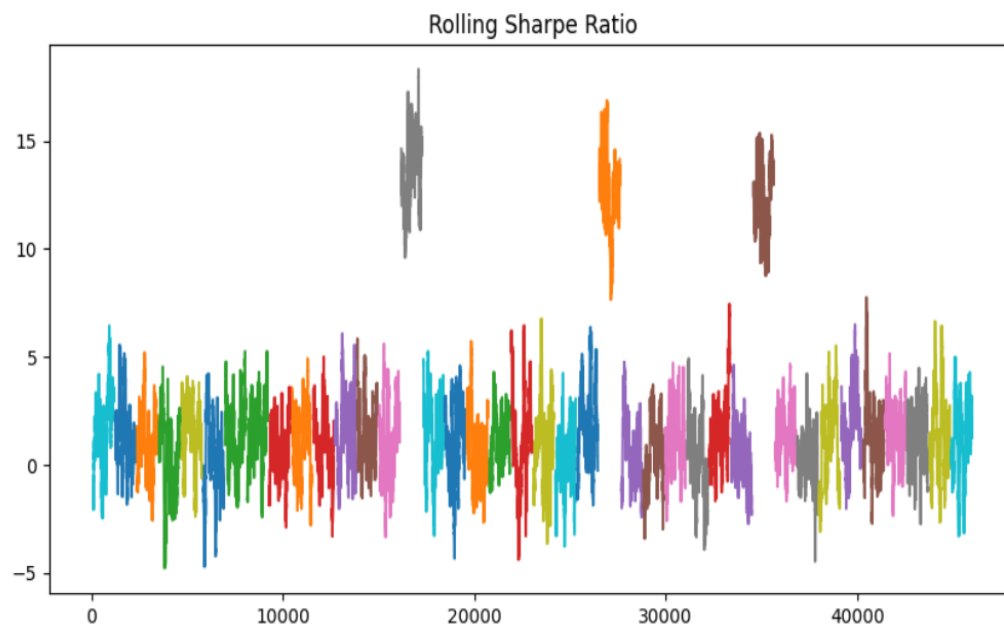
Based on the analysis:

- Conservative investors may consider Debt Funds.
- Moderate investors may choose Hybrid Funds.
- Aggressive investors may prefer Equity Funds.
- Investors should diversify across fund categories.
- Risk metrics should be considered alongside returns.

8. Limitations

- The analysis is based on historical mutual fund data and may not accurately predict future performance.
- Market conditions, economic changes, and external events can significantly impact fund returns.
- Some datasets may contain missing or incomplete information.
- Risk metrics such as VaR and CVaR are based on past trends and assumptions.
- The project does not consider investor-specific factors such as financial goals, investment horizon, or risk tolerance.
- Mutual fund performance can change over time, requiring periodic updates to the analysis.

Rolling Sharpe Ratio



Advanced Analytics

```
import pandas as pd

Click to add a breakpoint "data/raw/01_fund_master.csv"
nav = pd.read_csv("data/raw/02_nav_history (2).csv")
aum = pd.read_csv("data/raw/03_aum_by_fund_house.csv")
sip = pd.read_csv("data/raw/04_monthly_sip_inflows.csv")

print(fund.head())
```

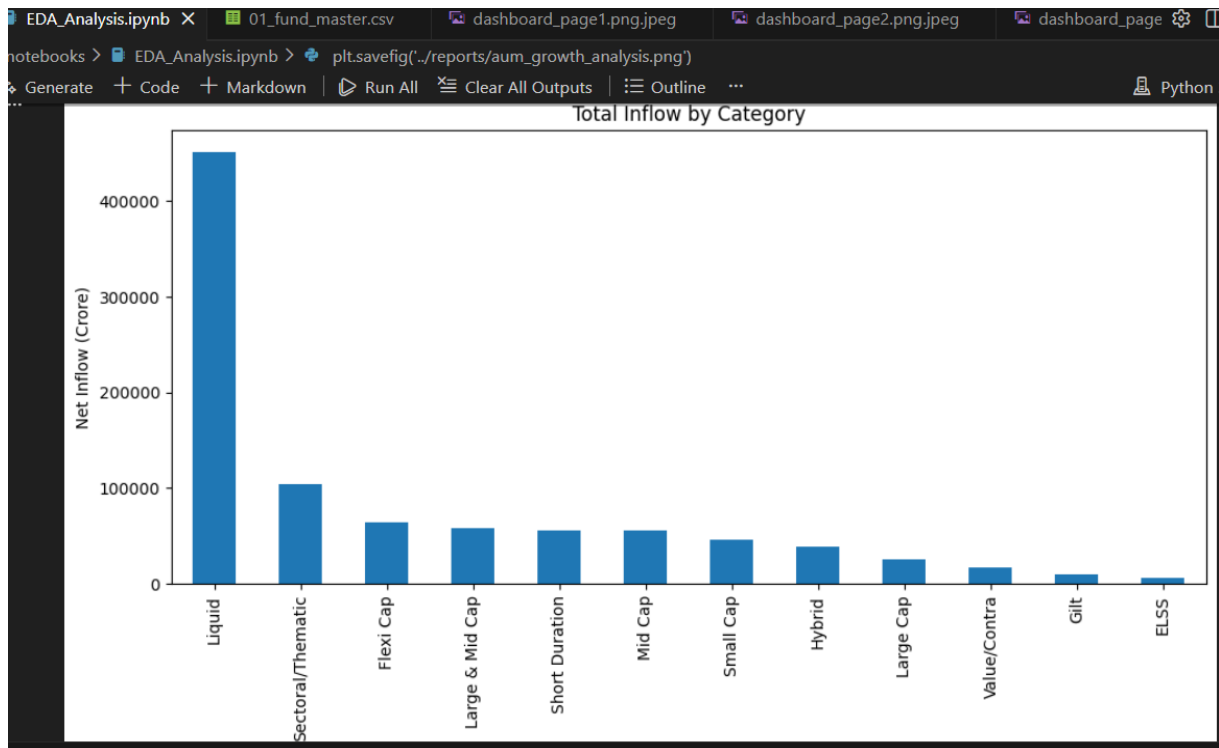
[12]

...	amfi_code	fund_house	scheme_name	\
0	119551	SBI Mutual Fund	SBI Bluechip Fund - Regular Plan - Growth	
1	119552	SBI Mutual Fund	SBI Bluechip Fund - Direct Plan - Growth	
2	119598	SBI Mutual Fund	SBI Small Cap Fund - Regular Plan - Growth	
3	119599	SBI Mutual Fund	SBI Small Cap Fund - Direct Plan - Growth	
4	119120	SBI Mutual Fund	SBI Magnum Gilt Fund - Regular Plan - Growth	

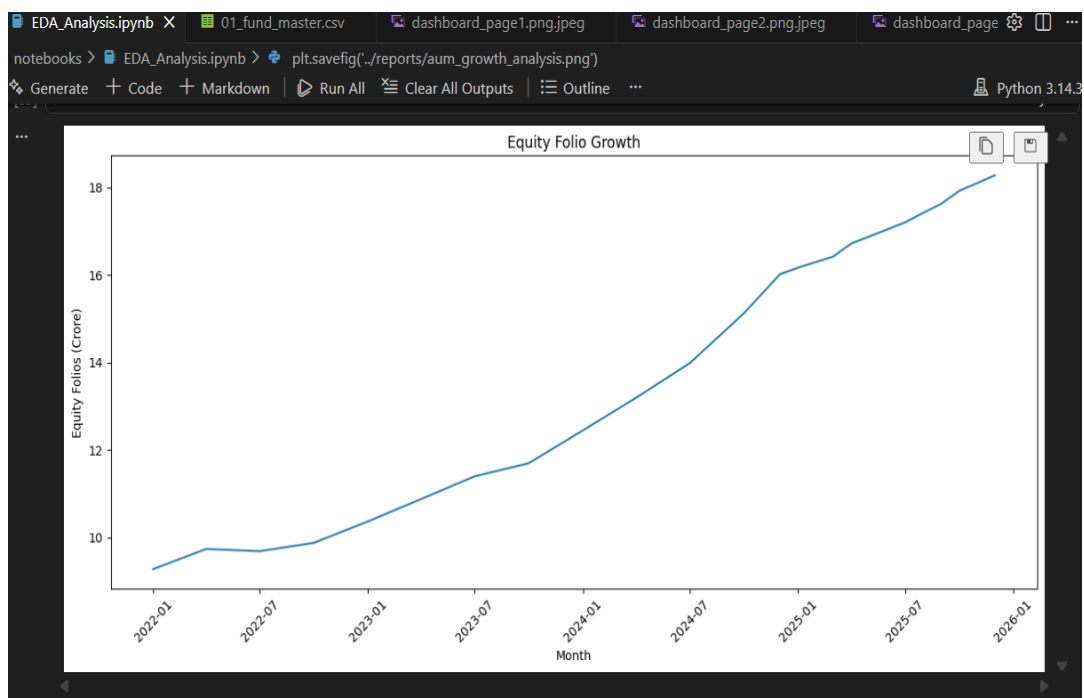
	category	sub_category	plan	launch_date	benchmark	\
0	Equity	Large Cap	Regular	2006-02-14	NIFTY 100 TRI	
1	Equity	Large Cap	Direct	2013-01-01	NIFTY 100 TRI	
2	Equity	Small Cap	Regular	2009-09-09	BSE 250 SmallCap TRI	
3	Equity	Small Cap	Direct	2013-01-01	BSE 250 SmallCap TRI	
4	Debt	Gilt	Regular	2000-12-30	CRISIL Dynamic Gilt Index	

	expense_ratio_pct	exit_load_pct	min_sip_amount	min_lumpsum_amount	\
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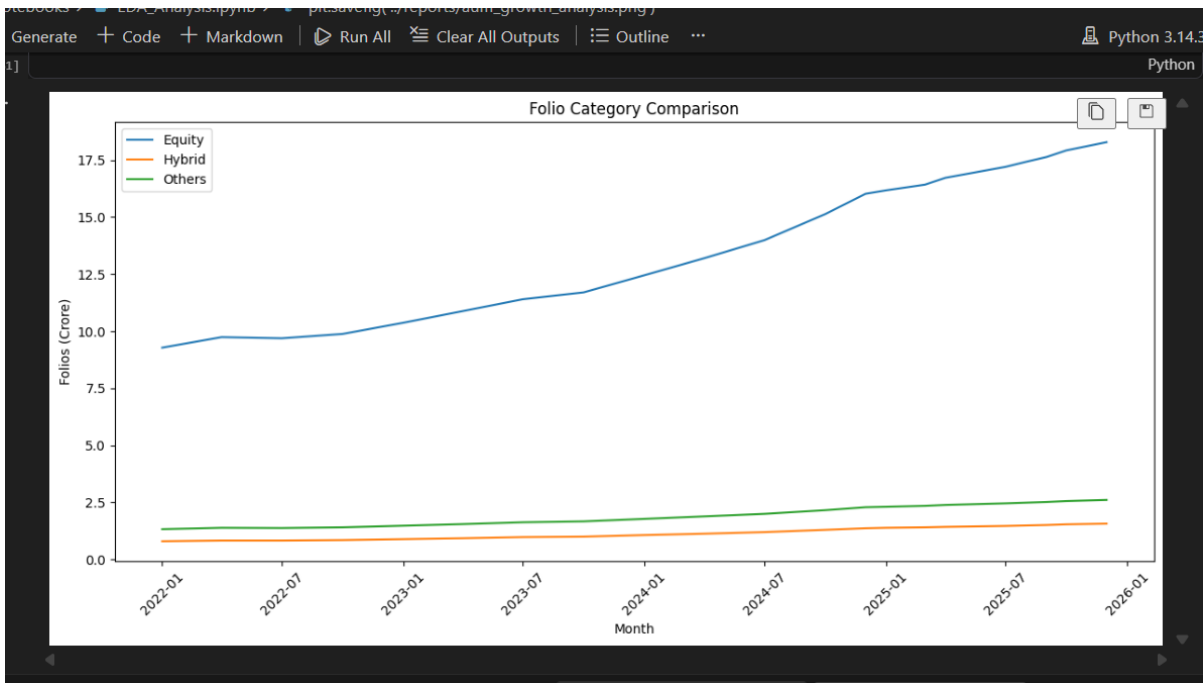
EDA Analysis



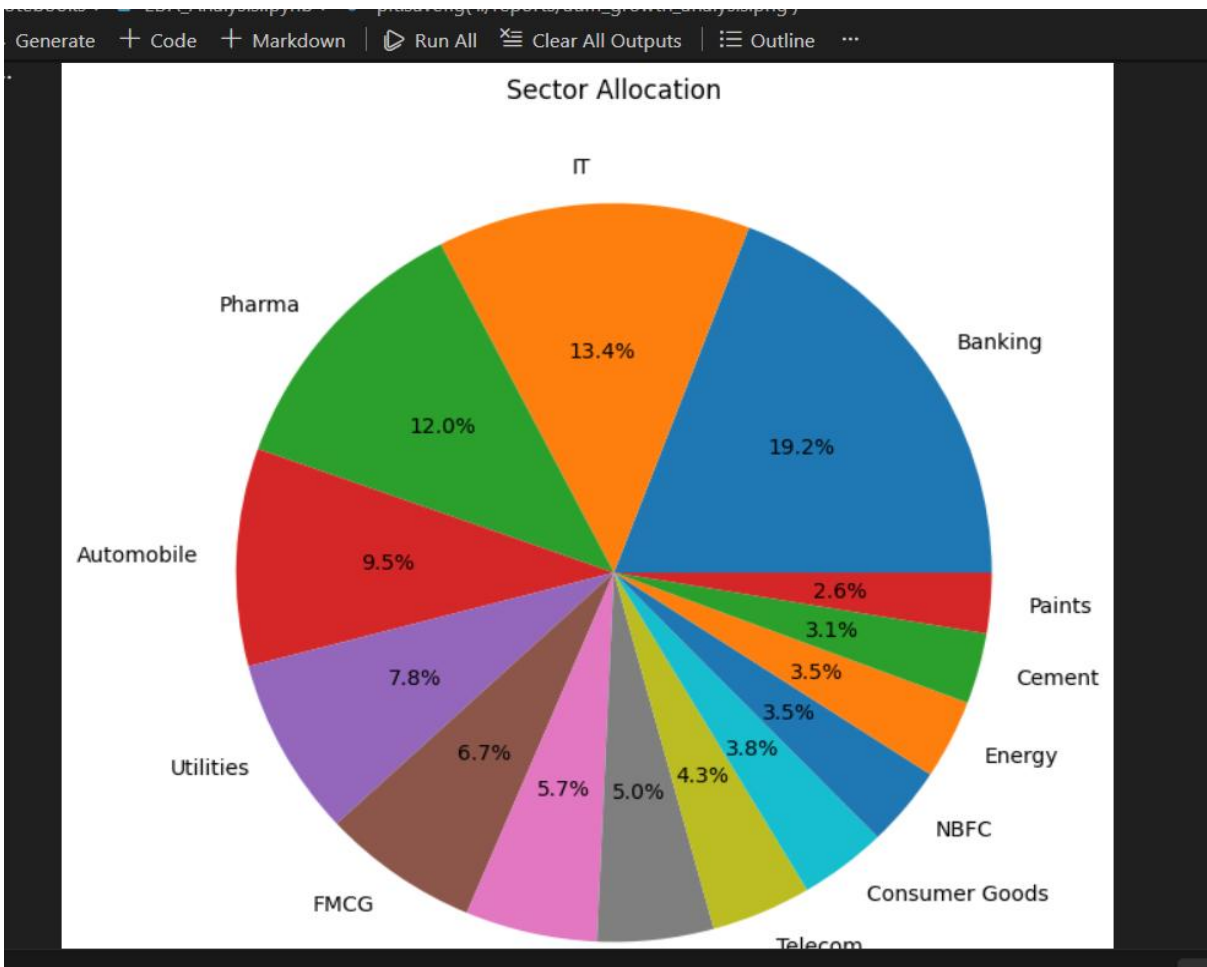
Equality Folio Growth



Folio Category Comparison



Sector Allocation



Appendix

Deliverables

- Advanced_Analytics.ipynb
- var_cvar_report.csv
- rolling_sharpe_chart.png
- recommender.py
- Power BI Dashboard
- Final Report PDF
- Presentation PPT

Tools Used

- Python
- Pandas
- NumPy
- Matplotlib
- Power BI
- GitHub

Thank You...